

Examiner reports

Q1.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there was an error, it was usually the addition of a function of e^x or a '+ c'.

Part (b) was answered very well with many of the candidates gaining full marks, although many then went on to try and simplify their expression, which caused a loss of marks in part (c).

Some attempts at the chain rule were seen, quite often resulting in an $\frac{x+1}{x}$ answer. Although

many correct responses were seen, part (c) was not answered as well as the two previous parts. Most of the errors came from following through from an incorrect expression in part (b). Some candidates found the gradient of the tangent and its equation but not the normal gradient. A number of candidates left their gradients in terms of x .

Q2.

Part (a)(i) was well answered by many candidates, who obtained full marks. Many often went on to obtain $e^x(x^2 + 2x - 3)$ which helped them later in the question. Part (a)(ii) was also well answered but not quite as well as part (a)(i). Many candidates obtained partial credit for reasonable attempts at the product rule but often there was a term missing.

In part (b)(i), the removal of e^x proved to be a major challenge: a large number of candidates chose to take logs (incorrectly) and, although some were lucky in the way they had written out the next step and recovered marks, many generated unmanageable expressions. In part (b)(ii), many correct responses were seen.

Q3.

- (a) This was well answered although common incorrect answers were $3\sec^2 x$ and $\sec^2 3x$.
- (b) Most candidates were able to apply the quotient rule but a large number of candidates were unable to correctly expand the numerator. A common response was $3(2x + 1) - 2(3x + 1) = 6x + 3 - 6x + 2$.

Q4.

Part (a) This was very well answered by the majority of candidates. Where errors occurred it was where candidates left their answer in terms of u or failed to multiply by the derivative of $3x$.

Part (b) This question proved difficult for many of the candidates although fully correct responses were seen. Most were able to achieve part marks for $du/dx = 2$ and many realised they had to write the integral completely in terms of u . It was at this stage factors

of $1/2$ were missing or dx was incorrectly replaced by $2 du$. A few candidates who correctly evaluated the integral in terms of u failed to put the expression back in terms of x .

Q5.

Many candidates lost marks on this question from careless work or failure to write answers to the correct degree of accuracy.

Part (a) Although many candidates obtained the correct method mark

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$, a significant number who reached this stage were unable to cope with $x \times \frac{1}{x}$ and equated it to 0, giving an answer of $\ln x$.

Part (b) Many candidates failed to realise the connection with the previous part. A significant number of candidates just wrote down the correct answer with no method indicated.

Part (c) Many candidates scored the method mark; some for the correct substitution and many for following through their answer to (b). Some candidates lost the accuracy mark for not writing the answer in exact form but going straight from $(5\ln 5 - 5) - (1\ln 1 - 1)$ to 4.047. Surprisingly, a number of candidates could not evaluate $-5 - (-1)$ correctly.

Q6.

Part (a) Reasonable attempts were made by most candidates.

Part (b)(i) Many candidates answered this correctly but there were many errors such as dividing by 2 first or writing $\sin^{-1} 2x$ as $1/\sin 2x$

Part (b) (ii) This part was usually answered correctly. The usual error was the inclusion of a negative sign.

In part (c) although some candidates gave fully correct responses the majority were

only able to do the dv 2 first part to get $\frac{dy}{dx} = \frac{2}{\cos y}$.

Q7.

Most candidates did well on this question, and even if some showed in part (a) that they apparently could not solve a quadratic equation, the implicit differentiation required in part (b) was generally done well.

Part (a) Most candidates substituted $x = 1$ correctly, and solved the resulting quadratic equation, usually by factorising, although some attempted a solution by non quadratic techniques, and usually made an error.

In Part (b) Many clear derivations were seen here, although candidates often dropped a mark through not equating their expression to zero during the derivation. The other relatively common error was in the product rule, where some candidates were not convincing in their use of signs and whether or not dy/dx was attached to a term.

Part (c) Most candidates successfully used the answer given to calculate the values of the derivatives, with some sign errors being made. Some used their own version of the derivative whereas some left the result in term of y , not realising they were to use the result of part (a).

Part (d) This part of the question defeated most candidates. Although most realised the derivative had to be set to zero, they could take it no further, and many who did write down $y - 6x = 0$ then couldn't see how to proceed. Some candidates actually "cross multiplied" to get $y - 6x = 2y - x$, usually before abandoning the attempt. Some candidates were convinced this part of the question was to do with the second derivative and did a great deal of work for no credit. Some used the given answer of $33x^2 - 5 = 0$ and solved it for x before abandoning or doing things such as substituting into the expression for the gradient.

Q8.

Part (a) was reasonably well answered although there were some candidates who integrated using the product rule. In part (b)(i) candidates were expected to use the chain rule, but some candidates tried to expand the brackets. This method could score full marks but candidates who used this approach, in general, made mistakes on the expansion. Part (b)(ii) used the previous part; however a large number of candidates tried to use integration by parts and found this difficult. Candidates should be guided to look for relationships between parts of a question.



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